

Nullnet encrypted endpoint lifecycle: before and after

23 September 2026 · Linux 6.12.95+deb13-amd64 · hosts 192.168.1.103 and .104. Each host: 8 vCPUs, about 16 GiB RAM, KVM, AMD Ryzen 7 7840HS.

The selected experimental design shares encrypted VXLAN sockets and reuses fully reset dedicated bridges. It keeps the current one-bridge-per-endpoint forwarding topology; bridge-netfilter settings stay unchanged. Same-host keeps per-edge MACsec. Cross-host keeps per-edge IPsec SAs and encrypted VXLAN metadata, using authenticated marks to share one UDP port. The selected cross-host variant uses unique request IDs, a shared inbound selector policy, per-edge outbound policies and a 4,096-packet replay window. A bounded pool holds anonymous, down, address-free dedicated bridges between leases. Every active endpoint still has its own bridge, MACsec or VXLAN, addresses and independent keys. Pool reset is included in endpoint teardown. Destroying the entire empty pool remains about 20 seconds per 1,000 bridges and is reported separately; it is not a solved fast-destruction path.

Result boundary. These are measured complete kernel endpoint sequences in a source-matched Python/Netlink harness, with real Docker namespaces and separate two-host packet proofs. They are not measurements of the full Nullnet RPC/database/eBPF/control-plane path. Product code was not changed. This establishes a tested implementation candidate; it does not establish production equivalence for every Nullnet recovery, routing or security path.

Unit. Every lifecycle trial has 1,000 endpoint halves on one host. Same-host therefore creates 500 encrypted pairs. A cross-host edge requires one endpoint on each host; timings from two isolated endpoint trials cannot be added or relabeled as measured complete distributed edges/s. The headline is comparable to the earlier 600+ bare-VXLAN endpoints/s measurement, not 1,000 full Nullnet networks/s.

Four-case result: Docker endpoints

Median seconds per 1,000 endpoints, three repeats per host. “Cold setup” includes batched discovery, pinned identity acquisition, host settings, bridge creation and crypto guards. “Final shutdown” also destroys the empty shared bridges.

Case / host	Before full	After endpoints	After cold / final	After endpoint rate/s
same-host setup · 103	26.604	1.546	1.755	647
same-host setup · 104	26.947	1.504	1.748	665
same-host teardown · 103	21.385	1.103	19.892	906
same-host teardown · 104	22.115	1.029	20.765	972
cross-host setup · 103	26.886	1.307	1.529	765
cross-host setup · 104	26.869	1.248	1.506	801
cross-host teardown · 103	40.836	1.142	20.008	876
cross-host teardown · 104	42.795	1.004	20.390	996

The approximately 1,000/s stretch target is approached for warm teardown, not uniformly for setup or complete cold lifecycle. Setup is around the earlier 600+ endpoints/s class. Count cold startup and final shutdown when that is the workload; keeping empty shared infrastructure is an explicit operating choice, not omitted endpoint cleanup.

Measurement and correctness boundaries

The before path reproduces current per-endpoint Docker inspection, namespace address/link subprocesses, native veth/bridge/VXLAN operations, stale-path probes, four crypto commands, salt hashing for IPsec, forwarding sysctl and FORWARD-policy updates. Teardown includes crypto cleanup, a link inventory, ownership regrouping and acknowledged grouped link deletion. Fixtures start clean: stale-path probes find nothing. Omitting repeated stale probes in the after path depends on authoritative lifecycle ownership and startup reconciliation; this benchmark does not establish the cost or correctness of migrating an already-dirty production host. It models current cleanup at 128 endpoints per idle batch. Same-host takes the complete per-edge lifecycle lock, including both halves' shared transport creation.

The selected after path keeps 32 bounded workers and eight CLI slots, namespace-bound Netlink sockets, native MACsec or XFRM/TC, one startup forwarding transaction, dedicated bridge leases and 256-endpoint idle deletion batches. Docker identity acquisition pins a network namespace FD and pidfd for each of 12 benchmark containers and checks process death before and after setup. Cross-host uses reserved UDP 4790; using a broad fail-closed rule on Docker Swarm's 4789 would be unsafe. The experiment uses dummy keys and addresses and does not read production keys.

Timing excludes creation of the 12 Docker fixtures, creating the isolated outer test namespace, debug inventories and post-run packet checks. It includes creation/removal of the endpoint's own objects. Standalone trials, where present, include each named endpoint namespace and default route. The steady setup clock starts after shared infrastructure is initialized; cold accounting adds all recorded initialization. Initializing the MACsec generic-Netlink family and interpreter overhead are not included in the cold formula; they are not production startup measurements.

Setup step tables report average elapsed latency spent inside each step per endpoint, including semaphore/RTNL waits. Parallel endpoint work overlaps: these rows must NOT be summed to get burst wall time. Full burst wall time is separately measured. Micro-operation tables separate subprocess execution from waiting for one of eight CLI slots. Teardown phases are sequential wall intervals and can be added; residual records scheduling, selection, assertions and loop overhead. These tables are not isolated single-operation capacity tests.

Network geometry: performance fixtures use an isolated dummy underlay and /30 addresses to measure lifecycle cost. Packet proofs use the real two-host underlay and /24 overlay subnets; same-host packet tests additionally exercise the production-style shared /29 with two endpoint/gateway addresses. Therefore 1,000 endpoints are not claimed to carry simultaneous end-to-end traffic during the throughput trial. MTU tests use endpoint MTU 1080. MACsec transport MTU in this harness is 1120.

Three repetitions characterize repeatability on these two hosts; they are not confidence intervals or a universal capacity promise. Python instrumentation and the interpreter contribute overhead. A Rust implementation may differ in either direction. Before and selected-after trials both use 32 workers / eight CLI slots. This is a native-configuration and topology comparison, not a one-line causal A/B.

Same-host setup: step breakdown

Step (ms per endpoint)	Before 103	Before 104	After 103	After 104
Docker PID inspection	79.359	79.350	—	—
Create/move endpoint veth	0.839	0.832	2.414	2.434
Namespace address (CLI)	72.904	72.846	—	—
Namespace link UP / native configuration	2.205	2.240	4.673	4.725
Gateway + dedicated bridge lease + endpoint attachment	0.966	0.979	6.696	6.653
Stale cross-host cleanup probe	71.572	72.535	—	—
Create/reuse encrypted transport veth	0.649	0.640	4.324	4.374
Install MACsec / IPsec + mark filters	78.227	78.814	12.128	11.711
Attach protected interface	0.400	0.381	2.658	2.601
Set IPv4 forwarding	73.920	75.327	—	—
Set FORWARD policy	186.599	188.488	—	—

Rows include waiting and overlap across endpoints. A dash means the work was removed, moved to initialization, or combined in another native step; it does not mean equivalent behavior may be skipped.

Measured full setup burst: 1,000 endpoints

Before · host 103		26.604 s
Before · host 104		26.947 s
After · host 103		1.546 s
After · host 104		1.504 s

Full setup seconds	Before	After warm	After cold
103	26.604 [26.550–26.817]	1.546 [1.459–1.605]	1.755 [1.663–1.875]
104	26.947 [26.690–27.000]	1.504 [1.441–1.583]	1.748 [1.675–1.846]

Same-host teardown: step breakdown

Seconds per complete 1,000-endpoint burst; median of three measurements.

Phase (seconds)	Before 103	Before 104	After 103	After 104
Crypto/stale state cleanup	0.231	0.220	0.015	0.015
Owned-link inventory	0.043	0.040	0.042	0.037
Assign retiring device group	0.072	0.067	0.065	0.058
Delete grouped links (ACK)	21.025	21.773	0.752	0.703
Remove final shared inbound policy	0.000	0.000	0.000	0.000
Reset bridge: DOWN, anonymous name, addresses/neighbors removed	0.000	0.000	0.225	0.211
Remove standalone namespaces	0.000	0.000	0.000	0.000
Other measured loop/check work	0.010	0.009	0.004	0.005
FULL ENDPOINT TEARDOWN	21.385	22.115	1.103	1.029
Destroy complete empty bridge pool	0.000	0.000	18.789	19.736
FULL FINAL SHUTDOWN	21.385	22.115	19.892	20.765

Medians of individual phases need not sum exactly to the median total. Crypto deletion is bounded concurrent work within its measured wall interval. No IPsec state is needed on the same-host MACsec path; the current stale-state probe still incurs process cost.

Measured full endpoint teardown: 1,000 endpoints



Cross-host setup: step breakdown

Step (ms per endpoint)	Before 103	Before 104	After 103	After 104
Docker PID inspection	206.014	204.013	—	—
Create/move endpoint veth	0.890	0.875	3.364	3.182
Namespace address (CLI)	219.584	218.131	—	—
Namespace link UP / native configuration	2.224	2.282	8.064	7.664
Gateway + dedicated bridge lease + endpoint attachment	0.944	0.962	12.062	11.527
Stale same-host lookup probes	0.125	0.139	—	—
Create/configure dedicated VXLAN	0.673	0.717	4.687	4.544
Key salt hashing subprocess	217.745	222.347	—	—
Install MACsec / IPsec + mark filters	6.212	6.377	10.181	9.786
Attach protected interface	0.001	0.001	3.066	2.938
Set IPv4 forwarding	1.744	1.747	—	—
Set FORWARD policy	186.973	186.406	—	—

Rows include waiting and overlap across endpoints. A dash means the work was removed, moved to initialization, or combined in another native step; it does not mean equivalent behavior may be skipped.

Measured full setup burst: 1,000 endpoints



Full setup seconds	Before	After warm	After cold
103	26.886 [26.559–26.965]	1.307 [1.287–1.327]	1.529 [1.490–1.566]
104	26.869 [26.865–27.000]	1.248 [1.206–1.301]	1.506 [1.450–1.556]

Cross-host teardown: step breakdown

Seconds per complete 1,000-endpoint burst; median of three measurements.

Phase (seconds)	Before 103	Before 104	After 103	After 104
Crypto/stale state cleanup	1.127	0.953	0.108	0.094
Owned-link inventory	0.035	0.030	0.032	0.030
Assign retiring device group	0.061	0.055	0.047	0.044
Delete grouped links (ACK)	39.607	41.755	0.735	0.622
Remove final shared inbound policy	0.000	0.000	0.002	0.002
Reset bridge: DOWN, anonymous name, addresses/neighbors removed	0.000	0.000	0.221	0.201
Remove standalone namespaces	0.000	0.000	0.000	0.000
Other measured loop/check work	0.005	0.005	0.003	0.003
FULL ENDPOINT TEARDOWN	40.836	42.795	1.142	1.004
Destroy complete empty bridge pool	0.000	0.000	18.895	19.401
FULL FINAL SHUTDOWN	40.836	42.795	20.008	20.390

Medians of individual phases need not sum exactly to the median total. Crypto deletion is bounded concurrent work within its measured wall interval. No IPsec state is needed on the same-host MACsec path; the current stale-state probe still incurs process cost.

Measured full endpoint teardown: 1,000 endpoints

Before · host 103		40.836 s
Before · host 104		42.795 s
After · host 103		1.142 s
After · host 104		1.004 s

Shared infrastructure and cold costs

Seconds added outside the warm endpoint setup timer. Per-container discovery is batched; namespace/pid FDs are held for the generation.

Initialization / shutdown	Same 103	Same 104	Cross 103	Cross 104
discovery_seconds	0.016	0.017	0.013	0.017
identity_init	0.000	0.000	0.000	0.000
pool_init	0.165	0.195	0.165	0.202
forwarding_init	0.028	0.036	0.029	0.028
security_init	0.000	0.000	0.004	0.004
shared_init	0.000	0.000	0.002	0.001
pool_finalize	18.789	19.736	18.895	19.401

Why the teardown was slow

The current topology creates one bridge per endpoint. Function-graph tracing and the Linux bridge source show a serial RCU barrier in bridge multicast destruction even when multicast snooping is disabled. This accounts for the roughly 21-second same-host teardown. Cross-host teardown adds a last-user UDP tunnel socket release for each unique destination port, producing roughly another 20 seconds. Replacing only IPsec or sharing only the VXLAN port leaves the bridge bottleneck.

A bounded pool of clean dedicated bridges removes bridge destruction from endpoint retirement; using marked IPsec lets distinct encrypted edges share a compatible UDP socket. The last shared socket and final bridge destructors still cost time and are explicitly included when the entire pool is removed.

Host 103: traced component deletion	Count	Seconds
bridge	16	0.338
veth	16	0.022
vxlan_shared	16	0.034
macsec	16	0.031

Host 104: traced component deletion	Count	Seconds
bridge	16	0.333
veth	16	0.022
vxlan_shared	16	0.034
macsec	16	0.031

Traced component runs use 16 objects, not 1,000; tracing is attribution evidence and adds overhead. Do not extrapolate their exact duration linearly or add inclusive nested function times.

[Bridge destructor source](#) · [VXLAN socket lifecycle](#)

Concurrent churn, isolation and reuse

Churn maintains 512 live endpoints, concurrently retires 256 and creates 256, and repeats six waves. IDs circulate through 1,024 slots with fresh incarnation keys. Each round checks exact device and crypto counts. Final checks require zero container peers and only clean anonymous bridge slots, without addresses, neighbors, dynamic FDB entries or multicast memberships. Both hosts and both selected encryption modes passed. This is mixed setup/teardown load, not 1,000/s independently for each direction.

Host / mode / batch	Mixed ops/s	p99 active ms	p99 completion ms	Max delete ACK ms
103 / same_macsec / 256	710	58.2	424.4	214.2
103 / marked_ipsec / 256	855	63.1	347.6	170.1
104 / same_macsec / 256	803	52.4	416.1	170.4
104 / marked_ipsec / 256	865	58.4	334.5	183.7

Active latency starts inside the worker; completion includes executor admission from the burst start. Delete-ACK duration is a userspace call duration, not a direct RTNL hold-time measurement. With 256 creates and 256 deletes per wave, each operation direction is half the mixed rate. The 256 idle batch is not automatically a safe replacement for the product's smaller batch while setups are active; retain bounded adaptive admission until integrated load testing establishes a tail-latency budget.

Bridge MACs are explicitly pinned and independently generated across hosts. Bridge reuse preserves the dedicated forwarding topology and the existing bridge-netfilter settings.

[Bridge MAC recalculation source](#)

Packet and lifecycle correctness evidence

The final marked-IPsec proof runs 32 Docker-backed edges on the two real hosts. Both directions pass all 32 connectivity tests; host gateway reachability and an unfragmented 1080-byte endpoint MTU pass. The dedicated-bridge candidate passes exact preservation of eight Ethernet frame formats, including customer tags, nested tags and priority bits, and the routed source-policy/SNAT/FORWARD-rule comparison under unchanged bridge-netfilter settings. Tagged-frame injection includes valid delivery controls; foreign and double-tagged probes deliver no marker to another endpoint. This checks carried Ethernet traffic, not a VLAN-based replacement for VXLAN.

The key negative test deliberately sends VNI 1 through edge 0's valid outbound SA. The receiving TC filter rejects the authenticated but mismatched edge mark; edge 0 survives, and restoring the correct mark restores edge 1. Independent wrong-key and missing-all-crypto tests fail closed. ESP replay beyond the selected 4,096-packet window increments `XfrmInStateSeqError`. Three fresh-key delete/recreate cycles retain connectivity and an unrelated edge survives each deletion.

Same-host tests exercise 64 endpoint halves / 32 pairs, bidirectional traffic, the shared /29 layout, MTU, TX-SA disable/restore, five fresh-key identifier/bridge/SCI reuse cycles and a real benchmark-container restart. The pinned old process generation is rejected before creating its veth; refreshing the namespace/pid FDs allows setup to recover. These passed on both hosts.

Evidence file: `dedicated-marked-proof.json`. Connectivity success counts: [32, 32].

Partial setup failure

Host	Mode	Observed result
103	same_macsec	Injected failure → owned cleanup → retry passed
103	marked_ipsec	Injected failure → owned cleanup → retry passed
104	same_macsec	Injected failure → owned cleanup → retry passed
104	marked_ipsec	Injected failure → owned cleanup → retry passed

The harness injects failure after partial crypto configuration, removes only the failed edge's objects, verifies unchanged survivor counts, and successfully reuses the slot. This proves the cleanup recipe, not automatic rollback in the unmodified Nullnet client.

Dataplane throughput check

Short single-edge iperf3 tests (5 seconds, one/eight TCP streams, both directions). These are sanity checks, not a sustained multicore or many-edge capacity claim.

Direction / streams	Current Gbit/s	Marked Gbit/s	Median change
forward / 1 streams	2.380 [2.341–2.381]	2.277 [2.043–2.354]	-4.3%
forward / 8 streams	2.385 [2.361–2.442]	2.340 [2.129–2.356]	-1.9%
reverse / 1 streams	2.405 [2.367–2.420]	2.342 [2.332–2.346]	-2.6%
reverse / 8 streams	2.424 [2.423–2.478]	2.221 [2.204–2.354]	-8.4%

Three repetitions per path, alternating order; cells are median [min–max]. The selected candidate's measured change is shown above; these short samples do not establish zero dataplane regression. All six repeated 100-byte UDP tests at 10 Mbit/s had zero loss. These repeated samples use the final shared-inbound, unique-reqid, 4,096-window configuration with 32 installed edges. The separate 1,000-edge check below exercises a larger policy table; sustained traffic on all edges simultaneously remains an integration requirement.

Final combined cross-host configuration

The final design keeps per-edge SAs/keys and outbound marked policies, shares only the peer/port inbound policy, gives each edge a unique reqid, and enables a 4,096-packet replay window. At 1,000 edges this uses 2,000 independent SAs and 1,001 policies per host. Its separate full security proof rechecks old-frame rejection after advancing beyond that window, wrong SA/VNI, wrong key, missing crypto and fresh-key reuse.

Edge / direction / streams	Final Gbit/s	TCP retransmits
0 / forward / 1	0.846	33
0 / forward / 8	0.849	1565
0 / reverse / 1	0.921	182
0 / reverse / 8	0.892	1315
999 / forward / 1	1.686	73
999 / forward / 8	1.959	2867
999 / reverse / 1	1.908	6
999 / reverse / 8	1.983	5886

Final cumulative XFRM counters on hosts 103/104: sequence rejects ['0', '0'], missing states ['0', '974']. Counters cover installation and traffic.

Counter snapshot	Missing SA 103	Missing SA 104	Sequence 103	Sequence 104
before_install	0	0	0	0
after_install	0	974	0	0
after_pings	0	974	0	0
after_traffic	0	974	0	0

Missing-SA counts above arise during concurrent installation, before both hosts finish setup. The post-install count remains unchanged through all 2,000 pings and eight TCP samples, with no sequence errors. Product publication must wait for both endpoint acknowledgements. Full 1,000-edge connectivity passed in both directions. Residual edge-position-dependent throughput and TCP retransmissions are reported rather than hidden; this is not a claim of constant dataplane cost at arbitrary scale.

Standalone namespaces: complete additional sequences

These trials create and remove one named network namespace per endpoint and install its default route. Endpoint namespace lifecycle is included, unlike the Docker case. Gateway ping and default-route checks run after timing; namespace paths must be absent after cleanup. Three repeats, 1,000 endpoints per host.

Topology / host	Before setup s	After cold s	Before teardown s	After endpoint teardown s	After final s
same / 103	26.245	3.121	22.807	2.105	21.291
same / 104	26.249	3.104	22.862	2.115	21.624
cross / 103	26.263	2.367	42.371	2.033	21.105
cross / 104	26.114	2.507	43.799	1.779	21.496

same-host namespace contribution	Create mean ms/endpoint	Remove full wall s
before / 103	66.543	0.419
before / 104	65.142	0.445
after / 103	4.887	0.078
after / 104	4.869	0.081

cross-host namespace contribution	Create mean ms/endpoint	Remove full wall s
before / 103	191.649	0.413
before / 104	188.844	0.440
after / 103	7.868	0.090
after / 104	8.670	0.063

Standalone remains below 600 endpoints/s in these complete sequences. The selected standalone candidate also replaces ip netns add/delete with dedicated-thread unshare, bind mount, namespace restore, unmount and unlink. It creates fresh namespaces; it does not recycle a previously used isolation domain. Removing those processes reduces deletion overhead, but fresh namespace creation and cross-namespace device cleanup still cost enough to miss the target. Do not substitute the faster Docker numbers for this full sequence.

Standalone same-host: every setup step

Mean elapsed milliseconds per endpoint, including waits. These concurrent intervals do not add to burst wall time.

Step (ms/endpoint)	Before 103	Before 104	After 103	After 104
Create named network namespace	66.543	65.142	4.887	4.869
Create/move endpoint veth	1.019	0.987	3.126	3.135
Namespace address (CLI)	76.025	74.376	0.000	0.000
Namespace link UP / native configuration	2.177	2.171	7.334	7.546
Standalone default route	2.125	2.133	0.000	0.000
Gateway + dedicated bridge lease + endpoint attachment	1.309	1.310	8.719	9.081
Stale cross-host cleanup probe	71.911	70.982	0.000	0.000
Create/reuse encrypted transport veth	0.838	0.788	6.007	6.070
Install MACsec / IPsec + mark filters	74.624	77.588	28.811	26.567
Attach protected interface	0.569	0.572	3.898	3.868
Set IPv4 forwarding	72.392	73.988	0.000	0.000
Set FORWARD policy	190.959	191.416	0.000	0.000

Standalone same-host: every teardown phase

Sequential wall seconds per 1,000 endpoints.

Phase (seconds)	Before 103	Before 104	After 103	After 104
crypto_remove	0.242	0.218	0.016	0.017
link_dump	0.045	0.042	0.052	0.044
regroup	0.199	0.191	0.158	0.154
link_delete	21.881	21.945	1.443	1.476
last_peer_crypto	0.000	0.000	0.000	0.000
bridge_reset	0.000	0.000	0.359	0.334
namespace_remove	0.419	0.445	0.078	0.081
residual	0.014	0.014	0.011	0.010
endpoint_total	22.807	22.862	2.105	2.115
pool_finalize	0.000	0.000	19.153	19.489
final_total	22.807	22.862	21.291	21.624

Standalone cross-host: every setup step

Mean elapsed milliseconds per endpoint, including waits. These concurrent intervals do not add to burst wall time.

Step (ms/endpoint)	Before 103	Before 104	After 103	After 104
Create named network namespace	191.649	188.844	7.868	8.670
Create/move endpoint veth	1.122	1.094	4.954	5.114
Namespace address (CLI)	213.490	217.299	0.000	0.000
Namespace link UP / native configuration	2.234	2.272	13.657	14.518
Standalone default route	2.159	2.178	0.000	0.000
Gateway + dedicated bridge lease + endpoint attachment	1.323	1.317	16.688	17.831
Stale same-host lookup probes	0.210	0.219	0.000	0.000
Create/configure dedicated VXLAN	0.940	0.966	6.566	6.869
Key salt hashing subprocess	217.126	216.232	0.000	0.000
Install MACsec / IPsec + mark filters	6.046	6.097	14.057	15.122
Attach protected interface	0.001	0.001	4.433	4.807
Set IPv4 forwarding	1.700	1.685	0.000	0.000
Set FORWARD policy	189.187	187.488	0.000	0.000

Standalone cross-host: every teardown phase

Sequential wall seconds per 1,000 endpoints.

Phase (seconds)	Before 103	Before 104	After 103	After 104
crypto_remove	1.143	0.929	0.085	0.086
link_dump	0.037	0.035	0.035	0.035
regroup	0.151	0.147	0.120	0.105
link_delete	40.627	42.196	1.271	1.140
last_peer_crypto	0.000	0.000	0.002	0.002
bridge_reset	0.000	0.000	0.380	0.342
namespace_remove	0.413	0.440	0.090	0.063
residual	0.012	0.011	0.011	0.010
endpoint_total	42.371	43.799	2.033	1.779
pool_finalize	0.000	0.000	19.142	19.717
final_total	42.371	43.799	21.105	21.496

Candidate choice and remaining product work

Select a bounded pool of reset dedicated bridges with stable, host-unique MACs; native namespace and crypto configuration; generation-bound Docker namespace handles; initialization/reconciliation of global forwarding; and shared-port per-edge marked IPsec cross-host. Same-host retains MACsec. Attach a protected transport to its dedicated bridge only after crypto and inbound identity filters are installed. Return a bridge lease only after acknowledged link removal and full reset. Keep pool destruction separate and explicit; it still exceeds the requested rate.

The endpoint candidate passed the packet and lifecycle checks described above. The remaining work is product integration and its four project gates: authoritative Docker generation/event reconciliation, cancellation and crash recovery, adoption of legacy bridges into the pool, per-edge locks, ownership-aware purge/discovery, lease generations and named bridge routes, ciphertext-only eBPF admission, reserved-port and mark allocation, firewall reload repair, key rotation/nonce lifetime, and real encrypted cold/warm concurrent Nullnet load including /api/graph state. No full CI or product multi-host regression claim is made for these report-only prototype files.

The final read-only audit on both hosts also found unchanged original link identities/MTUs/MACs and routes, zero benchmark containers/namespaces and zero temporary test-peer map entries. The existing application containers were left running. Test suites assert unchanged original Docker PID/start/network state. Packet work temporarily admits only the two isolated benchmark underlay IPs through the existing known-peer map and removes those exact entries afterward. No production firewall flush, application restart, product commit or deployment was performed.

The proposal-by-proposal rationale is in `change-rationale.md` / `change-rationale.pdf`, with individual documents in `changes/`. It states what changes, why equivalence is plausible, which tests passed, and which implementation obligations remain. A passing packet test is not a proof of every possible security property.

Retained evidence and reproducibility

Raw trial JSON and traces are retained per host in compressed archives. `sequence-summary.csv`, `step-breakdown.csv`, `operation-breakdown.csv` and `all-trials.csv` expose the numeric data. `lifecycle.py` and `native_*.py` are Linux experimental harnesses, not product libraries. `run-suite.py` creates only labeled benchmark containers and namespaces; `traffic-node.py` and the proof coordinators require the documented lab topology and temporary exact peer-map admission. Do not run their hardcoded lab credentials or addresses on another environment. Use `dedicated-verified-*` for selected Docker results and `dedicated-standalone-verified-*` for standalone results.

Appendix: low-level setup operations

For each operation, values are mean active call milliseconds per endpoint, then median across three runs. Netlink calls include kernel/RTNL wait. CLI execution excludes semaphore queue time; a separate queue total follows. Socket setup, hashing in Python and scheduler overhead live in the enclosing step totals.

Same-host before

Operation	Calls/endpoint	Host 103 ms	Host 104 ms
CLI MACsec RX SA	1	1.341	1.348
CLI MACsec RX channel	1	1.338	1.349
CLI MACsec TX SA	1	1.398	1.397
CLI MACsec device create	1	1.679	1.669
CLI docker inspect	1	11.227	11.263
CLI ip xfrm state deleteall	1	1.471	1.443
CLI iptables -P FORWARD ACCEPT	1	186.575	188.463
CLI namespace address	1	2.352	2.388
CLI namespace link UP	1	2.189	2.222
CLI semaphore queue (all commands)	0	352.892	358.832
CLI sysctl -w net.ipv4.ip_forward=1	1	1.904	1.879
MACsec configure	1	0.253	0.244
MACsec lookup	2	0.182	0.168
VXLAN lookup	1	0.083	0.089
endpoint veth configure	1	0.204	0.202
endpoint veth create	1	0.670	0.663
endpoint veth lookup	2	0.209	0.199
gateway address	1	0.081	0.075
gateway configure	1	0.109	0.110
gateway create	1	0.318	0.309
gateway lookup	2	0.180	0.189
transport veth configure	1	0.099	0.093
transport veth create	0.5	0.322	0.331
transport veth lookup	2	0.194	0.195

Same-host after

Operation	Calls/endpoint	Host 103 ms	Host 104 ms
MACsec RX SA	1	2.351	2.391
MACsec RX channel	1	2.436	2.432
MACsec TX SA	1	3.301	3.154
MACsec configure	1	1.411	1.353
MACsec create	1	1.281	1.289
MACsec lookup	2	2.439	2.450
endpoint veth address	1	1.365	1.355

Operation	Calls/endpoint	Host 103 ms	Host 104 ms
endpoint veth configure	2	2.808	2.801
endpoint veth create	1	2.233	2.242
endpoint veth lookup	1	1.768	1.774
gateway configure	1	1.419	1.388
transport veth address	1	1.032	1.091
transport veth configure	2	2.497	2.565
transport veth create	0.5	0.822	0.814
transport veth lookup	4	4.835	4.867

Cross-host before

Operation	Calls/endpoint	Host 103 ms	Host 104 ms
CLI docker inspect	1	11.381	11.419
CLI ip xfrm policy add	2	3.017	3.078
CLI ip xfrm state add	2	3.135	3.227
CLI iptables -P FORWARD ACCEPT	1	186.948	186.380
CLI namespace address	1	2.506	2.542
CLI namespace link UP	1	2.207	2.264
CLI semaphore queue (all commands)	0	629.898	630.015
CLI sha256sum	1	3.102	3.153
CLI sysctl -w net.ipv4.ip_forward=1	1	1.726	1.728
MACsec lookup	2	0.098	0.104
VXLAN configure	1	0.362	0.384
VXLAN create	1	0.236	0.253
VXLAN lookup	2	0.052	0.054
endpoint veth configure	1	0.197	0.206
endpoint veth create	1	0.713	0.703
endpoint veth lookup	2	0.201	0.183
gateway address	1	0.071	0.073
gateway configure	1	0.110	0.108
gateway create	1	0.306	0.308
gateway lookup	2	0.169	0.178
transport veth lookup	1	0.019	0.022

Cross-host after

Operation	Calls/endpoint	Host 103 ms	Host 104 ms
IPsec policy outadd	1	0.132	0.101
IPsec state inadd	1	0.103	0.095
IPsec state outadd	1	0.127	0.107

Operation	Calls/endpoint	Host 103 ms	Host 104 ms
TC inbound accept create	1	2.151	2.153
TC inbound drop create	1	2.380	2.343
TC outbound mark create	1	2.523	2.495
TC qdisc create	1	2.488	2.318
VXLAN configure	1	3.062	2.934
VXLAN create	1	1.924	1.868
VXLAN lookup	1	2.789	2.662
endpoint veth address	1	2.502	2.420
endpoint veth configure	2	5.015	4.721
endpoint veth create	1	3.088	2.878
endpoint veth lookup	1	2.403	2.358
gateway configure	1	2.615	2.558
transport veth address	1	1.995	1.869
transport veth configure	1	2.080	1.985
transport veth lookup	1	2.557	2.463